Computational experiment: does an interface perturbation create a decaying charge profile?

Status: completed numerical model experiment. **Not a laboratory experiment and not a material-specific prediction.** No nickelate sample was grown or measured. No Google Colab runtime was connected in this session. The notebook below is Colab-compatible; this computation was executed with Python 3, NumPy and Matplotlib in the sandbox.

Hypothesis tested

The proposal's narrow hypothesis was that an interface perturbation in an infinite-layer nickelate heterostructure produces a layer-resolved electronic reconstruction, strongest at the interface and decaying into the film. The specific model-level prediction tested here was an approximately exponential layer profile, $\Delta n_j \propto \exp(-j/\xi)$.

Null and alternatives

- **H_1-model:** An interface perturbation changes the layer occupations, with the response decaying approximately exponentially away from the interface.
- **H_0-model:** The perturbation produces no detectable charge redistribution, or the redistribution is not a simple interface-localized exponential profile.

Software and model

A one-orbital, non-interacting tight-binding slab was diagonalized on a 45×45 in-plane momentum grid. The slab contains 4, 6, 8, or 12 equivalent layers. Total electron filling was fixed at 0.70, 1.00, or 1.30 electrons per site including spin. Energies are in units of the in-plane hopping t, with illustrative interlayer hopping t_perp/t = 0.20 and temperature T/t = 0.04. None of these values is fitted to nickelate measurements.

At each thickness and filling, we compared a reference slab with: (1) a positive or negative onsite potential of ±0.30t on the interface layer; (2) reduced hopping on the first interlayer bond, from 0.20t to 0.05t; (3) a +0.15t potential on the first two layers, used only as a simple defect/electrostatic proxy; (4) a uniform 5% hopping increase, used only as a strain proxy; and (5) the interface potential combined with the uniform hopping change. The total film filling was re-equilibrated to its fixed target for each case. Thus, the imposed interface potential can redistribute charge between layers but cannot change the film's total electron count.

The calculation has no substrate reservoir, electrostatic Hartree screening, Sr chemistry, oxygen vacancies, multiple Ni orbitals, spin, electronic correlations, pairing interaction, superconducting order parameter, resistance, magnetic response, or critical temperature. “Vacancy proxy” and “strain proxy” are deliberately not literal chemistry or elastic-strain models.

Results

For the representative case of 8 layers and filling 1.00, the +0.30t interface potential changed the interface-layer occupation by **−0.1261 electrons/site**. The −0.30t potential reversed the sign, giving **+0.1258 electrons/site**. The net occupation change summed over all eight layers was numerically zero (within 10^{-15}), as required by the fixed-total-filling setup. The first-layer change therefore was compensated across the other layers.

The profile did **not** support the proposed simple exponential localization in this model. For the +0.30t case, the log-linear exponential fit had $R^2 = 0.068$ at 8 layers. At 12 layers it had $R^2 = 0.034$. The 12-layer profile after the interface layer was positive and broadly distributed: values at layers 1–11 were approximately +0.0108 to +0.0166 electrons/site, rather than decaying steadily toward zero. Across the tested thicknesses, the exponential-fit R^2 remained low (about 0.03–0.13 for the positive interface perturbation at filling 1.00). Under this model, **the imposed perturbation caused redistribution, but the predicted exponential decay was unsupported**.

Other tests show how strongly the result depends on the assumed physical term. At 8 layers and filling 1.00, reducing only the first interlayer hopping to 0.05t changed the interface-layer occupation by about **−0.000009 electrons/site**, essentially zero in this calculation. The two-layer +0.15t potential proxy changed it by **−0.0545 electrons/site**. A uniform 5% hopping increase alone changed the interface-layer occupation by about **+0.000004 electrons/site**, also essentially zero at fixed filling. Applying the interface potential together with this hopping change gave **−0.1205 electrons/site**, close to the interface-only response. These are outcomes of the selected one-orbital model—not statements about actual interfaces, vacancies, strain, or superconductivity.

For the representative interface-potential case, increasing thickness from 4 to 12 layers changed the interface-layer occupation change from approximately −0.109 to −0.132 electrons/site, while the root-mean-square profile change fell from 0.063 to 0.040 electrons/site because the compensating charge was distributed over more layers. This is a finite-slab, fixed-charge effect, not a measured thickness trend in nickelates.

Numerical check

The interface-layer occupation change for the 8-layer, filling-1.00, +0.30t condition was −0.12643 at a 55×55 momentum grid, −0.12561 at 65×65, and −0.12582 at 75×75. The

corresponding RMS profile changes were 0.04782, 0.04752, and 0.04760. This indicates numerical stability to about 1% over the last grids for this one representative quantity. It does not validate the model's physical assumptions.

Interpretation

The model demonstrates a limited mathematical point: **an assumed interface-localized onsite energy can redistribute electron occupation among layers**. It does not show that an experimentally realized $NdNiO_2/SrTiO_3$ interface has that potential, that the same profile occurs in a multi-orbital correlated material, or that the redistribution helps superconductivity.

As coded, the model does not support the proposal's exponential-decay form. That is not a falsification of interface reconstruction in real nickelates, because the calculation omits key ingredients—including the substrate as an electronic reservoir, self-consistent electrostatics, and Ni multi-orbital correlations. It is a falsification only of the claimed exponential profile **within this minimal model and parameter range**. A more material-relevant next computation would require a measured atomic stack/termination and parameterized multi-orbital electronic structure, ideally compared against depth-resolved measurements. Predicting superconductivity would additionally require a justified pairing model; this calculation has none.

Reproduce it

- Open the included `nickelate_interface_model_colab.ipynb` in Google Colab after uploading the notebook file. It needs only NumPy and Matplotlib, which are typically available in Colab.
- The same code is in `interface_model_experiment.py`; running it writes `layer_charge_profiles.csv`, `model_summary.csv`, `k_grid_convergence.csv`, and `charge_reconstruction_profiles.png`.
- The figure and CSVs are model outputs, not experimental observations. Do not label them measured data or use them as evidence of a measured superconducting transition.